

What is Zooarchaeology?

Zooarchaeology is the study of the remains of animals from archaeological contexts. While there are many types of animal remains that can be preserved, the most common are bone (of vertebrate animals) and shell (of invertebrates). In rare cases, other animal tissues can be preserved, such as fish scales, bird feathers, and skin and hair of mammals. These differences in preservation are the result of **taphonomic** processes.

Zooarchaeology is the study of animal remains from archaeological contexts. While it can include the study of animal remains of any type, most zooarchaeologists study bone and shell, which are the most likely to be preserved archaeologically.

Taphonomy is the study of how organisms decay following death, especially once buried, and enter the archaeological record. Taphonomy includes the study of how organisms or their remains are disposed of by humans as well as how natural processes affect those remains once they are buried.

Humans modify bones in a variety of ways that contribute to their taphonomic preservation. When killing, butchering, and cooking an animal, one may damage various bones. In some cases, humans deliberately break bones before or after cooking to access bone marrow or enable the bones to fit within a pot. Eating can damage bones, as well; human teeth can leave marks on bone, though carnivores (such as dogs) and rodents (including mice and rats) that live with or around human settlements also chew on bones, damaging them in characteristic ways. Humans may also modify

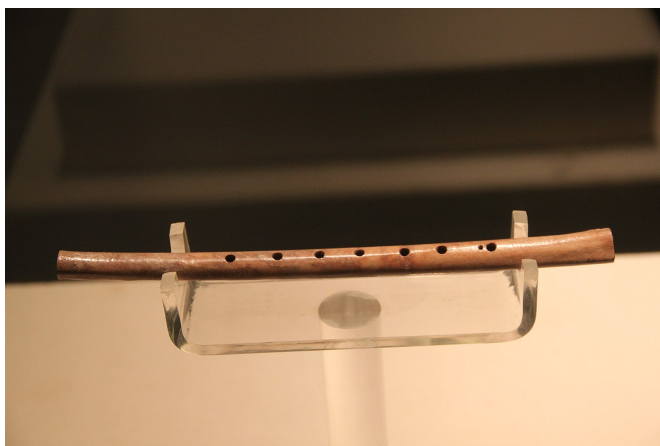
bones on purpose to make tools as diverse as carefully crafted bone arrowheads, needles, combs, burnishing tools, awls, and fishing hooks, as well as decorative items and even musical instruments!



A bone awl, used to punch holes in hide or leaves, or for other purposes. From New Zealand.

Source: [Wikimedia Commons](https://commons.wikimedia.org/wiki/File:Awl_bone,1933.379_(i).jpg)

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A bone flute from the Neolithic period of China.

Source: [Wikimedia Commons](https://commons.wikimedia.org/wiki/File:Early_Peiligang_Culture_Bone_Flute,_Wuyang.jpg)

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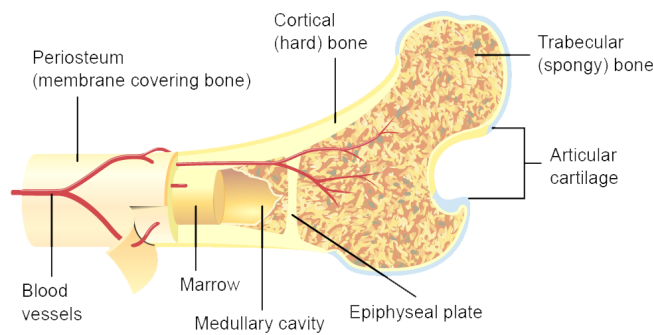
Zooarchaeology includes five stages of analysis:

- Recovery – Sampling bones and shell from an archaeological context
- Sorting – Separating like bones and shells from others
- Identification – Identifying each bone and observing damage or modification to it
- Quantification – Counting the remains in one of several ways

- Interpretation – Making inferences about past human behaviors and animal ecology from these results

Bone

Bone is the most ubiquitous animal material preserved in most types of archaeological sites. It is hard and durable archaeologically due to its structure, which combines the organic molecule **collagen**, a protein that gives bones their flexibility and resistance to fracturing, with inorganic **hydroxyapatite**, a mineral that combines calcium and phosphate and gives bone its strength and hardness. Bone comes in many forms, depending on the anatomy of the animal—from thin, hollow bird bones to the large, dense bones of heavy, terrestrial mammals. Bones include two types of tissue: **cortical bone**, which is hard and dense and typically present as the outer layer of bones, and **trabecular bone** (also termed “spongy bone”), which is spongy and often present in the **epiphyses**, or ends, of bones. In mammals, most bones that articulate (connect) with others have epiphyses that begin as cartilage when the animal is young. Those cartilaginous epiphyses then harden as the animal grows older and eventually fuse with the **diaphysis** (or shaft) of the bone. The age at which this occurs for each epiphysis of each bone is consistent from individual to individual, with some variation; so, the ages of juveniles who died before adulthood can be estimated by identifying which epiphyses had fused and which had not. Bone **marrow** is the fatty tissue that produces blood cells for the body, among other functions.



A diagram of bone in partial cross-section. Note the trabecular bone and the cortical bone, as well as the marrow within the medullary cavity.

Source: [Wikimedia Commons](https://commons.wikimedia.org/wiki/File:Bone_cross-section.svg)

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Collagen is the protein component of bone, found also in cartilage, which gives bone its flexibility.

Hydroxyapatite is the inorganic component of bone, composed of calcium and phosphate, which gives bone its strength.

Cortical bone is hard, dense bone, found on the outer surface of bones.

Trabecular bone is spongy, brittle bone, found in the ends or interior cavities of large bones.

Epiphyses are the ends of bones that articulate with other bones. Epiphyses form from cartilage as separate pieces of bone that eventually fuse with the **diaphysis** at an age determined by the developmental biology of the species.

The **diaphysis** is the shaft of a long bone. Diaphyses typically ossify (turn from cartilage to bone) earlier than the epiphyses with which they eventually fuse.

Bone marrow is a fatty tissue within bones that produces blood cells. It is typically found in the highest concentration within the medullary cavity of long bones, but is also present in **trabecular bone**.

Shell

Shell, the hard tissue of invertebrates, is durable and often well preserved. Some archaeological sites consist nearly entirely of shell; these are termed "shell middens." Shell produced by mollusks is composed of calcium carbonate, which hardens over an organic framework. These shells, especially those of the class Bivalvia (bivalves, including clams, mussels, oysters, etc.), can be large and durable enough to be common finds in archaeological contexts close to bodies of water with large populations of shellfish. Also relatively common are the shells of land snails (in class Gastropoda), which can be found in environments from forests to deserts.



A shell midden from coastal Argentina.

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